ALEXANDER JERSCHOW

Website
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US citizen
DOB: 10/28/2002, New York

EDUCATION

Math Master Nagoya University Immatriculated: Apr 2025
Advisor: Yoshimichi Ueda
Random matrices and neural networks
Japanese Instruction

Notable courses:

- Operator Theory (Yoshimichi Ueda, Syllabus)
- Topological K Theory (Chris Bourne, Syllabus)

Math Master 1 Semester University of Vienna Enrolled: Oct 2024-Feb 2025
GPA: 4 (=1 Austrian GPA scale)
English Instruction

Notable courses:

- Advanced Functional Analysis (Günther Hörmann, Syllabus)
- C* Algebras with Aspects in Quantum Physics (Günther Hörmann, Syllabus)
- Stochastic Analysis (Julio Backhoff, Syllabus)
- Stochastic Processes (Emmanuel Schertzer, Syllabus)
- Synthetic Geometry and Optimal Transport Seminar (Clemens Sämann, Roland Steinbauer, Michael Kunzinger, Chiara Rigoni, Syllabus)
- Advanced Partial Differential Equations (Jörg Weber, Syllabus)
- Measure and Integration Theory (Roland Zweimüller, Syllabus)

Math Bachelor with Distinction University of Vienna Enrolled: Sep 2021, Graduated: Feb 2024
GPA: 3.8 (=1.4 Austrian GPA scale)
Bachelor Thesis: Sanov's Inequality from an Information Theoretic Perspective
German Instruction

Notable courses:

- Topology and Functional Analysis (Gerald Teschl, Syllabus)
- Complex and Harmonic Analysis (Jose Luis Romero, Klaus Widmayer, Syllabus)
- Integration and Stochastics (Roland Zweimüller, Syllabus)
- Probability and Statistics (Mathias Beiglböck, Syllabus)
- Ordinary Differential Equations (Henk Bruin, Syllabus)
- Financial Mathematics (Julio Backhoff, Syllabus)
- Analysis III (Andreas Cap, Syllabus)
- Optimization and Modelling (Radu Ioan Boţ, Syllabus)
- Numerical Mathematics (Hermann Schichl, Norbert Mauser, Martin Ehler, Syllabus)
- Algebra II (Joachim Mahnkopf, Syllabus)

HONORS/AWARDS

- 2025 Merit Scholarship from the University of Vienna, funded by the Austrian Federal Ministry of Science, Research and Economics
- 2024 Merit Scholarship from the University of Vienna, funded by the Austrian Federal Ministry of Science, Research and Economics
- Ranking #3 in 2025 in the mathematics master's program at the University of Vienna
- Acceptance into the Presidential Honors Scholars Program (2020)
- Acceptance into the KUINEP program (2024)

COMPETITIONS

- Participant, William Lowell Putnam Mathematical Competition (Putnam)
- Participant, American Invitational Mathematics Examination (AIME)
- Participant, USA Computing Olympiad (USACO)

WORKS

- Bulk-edge sticking beyond the Perron mode in Gaussian softmax attention arXiv:2609.23949, Sep 2026
- Another Proof For the Existence of the Free Convolution Semigroup for Compactly Supported Measures, Oct 2025
- Combinatorial Definition of the R-transform and Voiculescu's Characterization, a Reference, Oct 2025
- Sanov's Inequality from an Information Theoretic Perspective (Bachelor's Thesis), Feb 2024

PROJECTS

- Heron — a flashcard website for language learning.

PRESENTATIONS/CONFERENCES

- Presentation at the 2026 Annual Japanese Junior Functional Analysis Conference, Yamaguchi, Japan, September 16-19, 2026 (conference website; slides in Japanese).
- Participated in KMI-Math School: Bridges Between Physics and Mathematics, Nagoya University, Japan, February 9-12, 2026.
- Presentation at 2025 Annual Japanese Junior Functional Analysis Conference (for master's students, PhDs and postdocs in Japan, website in Japanese). Presentation about Voiculescu's free R transform characterization via Nevanlinna Pick (slides in Japanese).
- Participated in organizing the Kyoto Random Matrix Theory Summer School 2025

TEACHING

- Complex Analysis (現代数学基礎 CIII, Fall 2025)
  Solutions (Japanese)
  Duties: providing Japanese solutions, presenting these Japanese solutions during recitation, Q&A, grading
  Professor: Hiroaki Kanno
- Foundations in Mathematics (数学演習 III, Spring 2025)
  Duties: giving hints to problems/answering questions during class, grading
  Professor: Kento Osuga

WIKIPEDIA CONTRIBUTIONS

Created:

- R-Transform (Pending Approval)
- Satz von F. und M. Riesz (F. and M. Riesz Theorem, German)
- 里斯兄弟定理 (F. and M. Riesz Theorem, Chinese Simplified)
- 里斯兄弟定理 (F. and M. Riesz Theorem, Chinese Traditional)
- Теорема плотности Капланского (Kaplansky Density Theorem, Russian)*
- Schattennorm (Schatten Norm, German)*
- Норма Шаттена (Schatten Norm, Russian)*

Contributed:

- Kaplansky Density Theorem
- シャッテンノルム (Schatten Norm, Japanese)
- Hilbert-Schmidt-Operator (Hilbert-Schmidt Operator, German)
- Оператор Гильберта-Шмидта (Hilbert-Schmidt Operator, Russian)
- Liste mathematischer Sätze (List of Theorems, German)
- 数学定理列表 (List of Theorems, Chinese Simplified)
- 數學定理列表 (List of Theorems, Chinese Traditional)

* Translations from other languages

FREE TIME

- Language: English (US citizen), fluent in German and Russian, upper intermediate-advanced in Japanese (I passed the highest level of the national Japanese Language Proficiency Test), beginner in Chinese
- Climbing, calisthenics, piano
- Member of the Nagoya University Gymnastics Club, as well as the Nagoya University Jazz Fusion Club